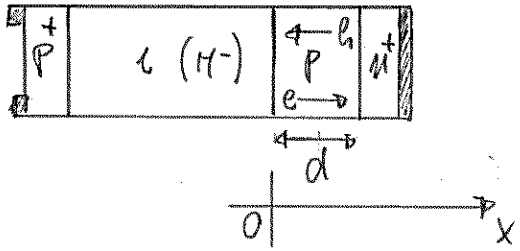
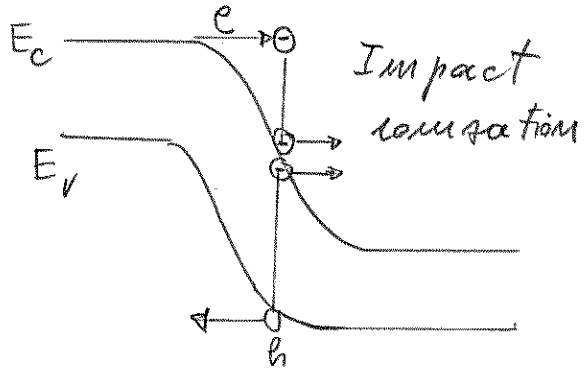
Avalanche Photodiodes



$$\begin{cases} \dfrac{di_e}{dx} = \alpha_e i_e + \alpha_h i_h \\ -\dfrac{di_h}{dx} = \alpha_e i_e + \alpha_h i_h \end{cases}$$

$$I = i_h(x) + i_e(x)$$
$$i_h(x) = I - i_e(x)$$



$$\frac{d\,i_e(x)}{dx} = \alpha_e i_e(x) + \alpha_h I - \alpha_h i_e(x)$$

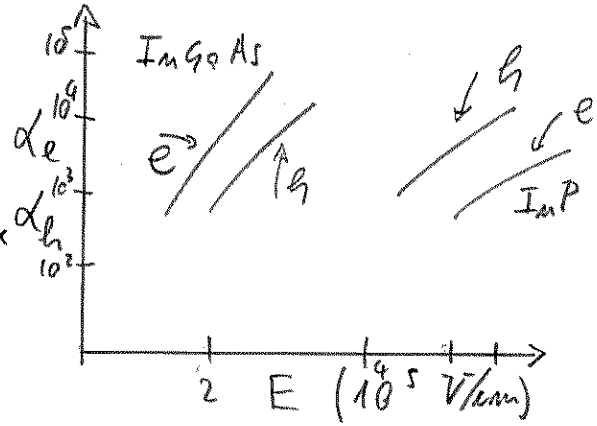
$$\frac{d\,i_e(x)}{dx} = (\alpha_e - \alpha_h)\, i_e(x) + \alpha_h I$$

Boundary cond.

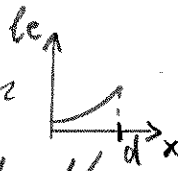
$$\begin{cases} I_h(d) = 0 \\ i_e(d) = I \end{cases} \qquad K_A = \frac{\alpha_h}{\alpha_e}$$

...

$$M = \frac{i_e(d)}{i_e(0)} = \frac{1 - K_A}{\exp[-(1-K_A)\alpha_e d] - K_A}$$



If $\alpha_h \simeq 0 \rightarrow K_A \simeq 0 \rightarrow M = \exp(\alpha_e d)$ first order process



If $\alpha_e \simeq \alpha_h \rightarrow K_A \simeq 1 \quad M = \frac{0}{0} \quad \lim_{K_A \to 1} M = \frac{1}{1 - \alpha_e d}$ $\begin{cases} d = 1/\alpha_e \\ M \to \infty \end{cases}$

$$M(f) = \frac{M_0}{\sqrt{1 + 4\pi^2 f^2 \tau_e^2 M_0^2}} \qquad \tau_e = C_A K_A \tau_{drift} \quad C_A \simeq 1$$

$$K_A << 1$$

InGaAs $\rightarrow K_A = \alpha_h/\alpha_e < 1$

InP $\rightarrow K_A = \alpha_e/\alpha_h < 1$

$$f_c = \frac{1}{2\pi \tau_e M_0} \simeq \frac{1}{2\pi\, \tau_{drift} K_A M_0}$$

$$R_{APD}(f) = M(f)\, R_\lambda = M(f)\, \frac{\eta e \lambda}{hc}$$

Noise in junction detectors

PIN

$$\sqrt{\langle i_n^2\rangle} = \sqrt{\left[2e\left(\cancel{\langle i_s\rangle} + \cancel{\langle i_d\rangle}\right) + 4KT/R_L\right]\Delta f}$$

$$SNR = \frac{i_s^{RMS}}{\sqrt{\langle i_n^2\rangle}} = \frac{R_\lambda^I \cdot P_{in}}{\sqrt{4KT/R_L\Delta f}} = \frac{R_\lambda^I P_{in}\sqrt{R_L}}{\sqrt{4KT\Delta f}} = P_{in}\sqrt{\frac{R_L}{4KT\Delta f}}\,\frac{\eta e\lambda}{hc}$$

$$NEP^*_{PIN} = \frac{NEP}{\sqrt{\Delta f}} = \frac{\sqrt{\langle i_n^2\rangle}}{R_\lambda\sqrt{\Delta f}} = \frac{hc}{\eta e\lambda}\cdot\sqrt{\frac{4KT}{R_L}}$$

APD

$$\sqrt{\langle i_n^2\rangle} = \sqrt{\left[2eM^2\left(\langle i_s\rangle + \langle i_d\rangle\right)F_A + 4KT/R_L\right]\Delta f}$$

$$F_A(M) = K_A M + (1-K_A)\left(2-\frac{1}{M}\right)$$

If $K_A \ll 1$

$$F_A(M) = 2 - \frac{1}{M} \simeq 2$$

$$NF = 10\log_{10} F_A = 3\,dB$$

$$SNR = \frac{i^{RMS}}{\sqrt{\langle i_n^2\rangle}} = \frac{M R_\lambda^I P_{in}}{\sqrt{(4KT/R_L)\Delta f}} = M\cdot P_{in}\sqrt{\frac{R_L}{4KT\Delta f}}\cdot\frac{\eta e\lambda}{hc} = M\,(SNR)_{PIN}$$

$$NEP^*_{APD} = \frac{1}{M\cdot R_\lambda^I}\sqrt{\frac{4KT}{R_L}} = \frac{1}{M}\cdot\frac{hc}{\eta e\lambda}\sqrt{\frac{4KT}{R_L}} = \frac{1}{M}\left(NEP^*\right)_{PIN}$$

If shot noise can be neglected, APDs give an improvement of SNR and NEP* by a factor M with respect to PIN

This is true in low light condition <u>only</u>

$$\eta_g = \frac{M i}{2\tau_c}\ [s^{-1}m^{-3}]$$

$$R_g = \eta_g \cdot \text{Volume}$$

$$M i_{s1} = 1.45\,10^{16}\cdot m^{-3}$$

$$\tau_c \simeq 10\,ms$$

$$\Rightarrow \eta_g \simeq 7.2\cdot 10^{17}\,s^{-1}m^{-3}$$

$$A = 40\times 40\ \mu m^2$$

$$W_d \simeq 1\ \mu m$$

$$Vol. \simeq 1.5\cdot 10^{-15}\,m^3$$

$$\Rightarrow R_g \simeq 10^3\ s^{-1}$$

@ 300 K

$$e\langle i_d\rangle = R_g e^2 \simeq 10^{-35}\,A\cdot C$$

$$R_L = 50\,\Omega$$

$$KT = 25\,meV$$

$$4KT/R_L \simeq 10^{-22}\,A\cdot C$$

Coherent detection → Homodyne

$$\mathcal{E}_c(t) = E_c \cos[\omega_c t + \vartheta(t)] \qquad \mathcal{E}_{lo} = E_{lo} \cos(\omega_c t + \varphi)$$

$$\mathcal{E}_{coh}(t) = \frac{1}{\sqrt{2}} \left\{ E_c \cos[\omega_c t + \vartheta(t)] + E_{lo} \cos(\omega_c t + \varphi) \right\}$$

$$I(t) = \varepsilon_0 c \, \mathcal{E}^2(t) \qquad i(t) = A_d R_\lambda I(t)$$

↑ light intensity ↑ Output current

$$i(t) = \frac{1}{2} \varepsilon_0 c A_d R_\lambda \left\{ E_c \cos[\omega_c t + \vartheta(t)] + E_{lo} \cos(\omega_c t + \varphi) \right\}^2$$

...

$$i(t) = \frac{1}{4} \varepsilon_0 c A_d R_\lambda \Big\{ \cancel{E_c^2} + \cancel{E_{lo}^2} + 2 E_{lo} E_c \cos[\vartheta(t) - \varphi] + \cancel{2 E_{lo} E_c \cos[2\omega_c t + \vartheta(t) + \varphi]}$$
$$+ \cancel{E_c^2 \cos[2\omega_c t + 2\vartheta(t)]} + \cancel{E_{lo}^2 \cos[2\omega_c t + 2\varphi]}$$

$$i(t) = \frac{1}{2} \varepsilon_0 c A_d R_\lambda E_{lo} E_c \cos[\vartheta(t) - \varphi]$$

$$i(t) = R_\lambda \sqrt{P_c P_{lo}} \cos[\vartheta(t) - \varphi]$$

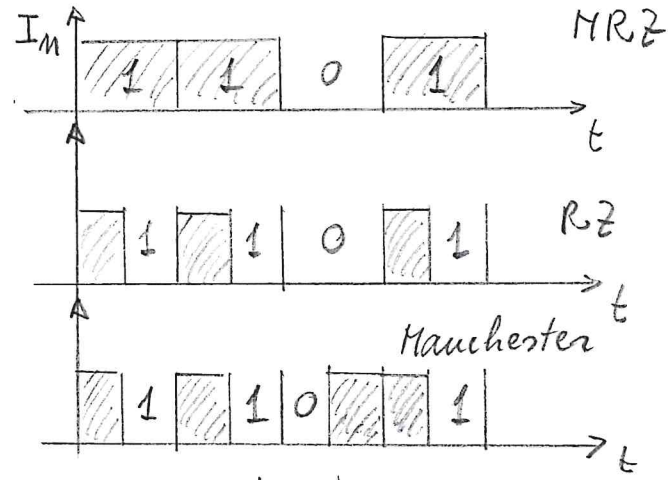
$$\vartheta(t) = \vartheta_0 + \psi(t) \qquad i(t) = R_\lambda \sqrt{P_c P_{lo}} \cos[\vartheta_0 - \varphi + \psi(t)]$$

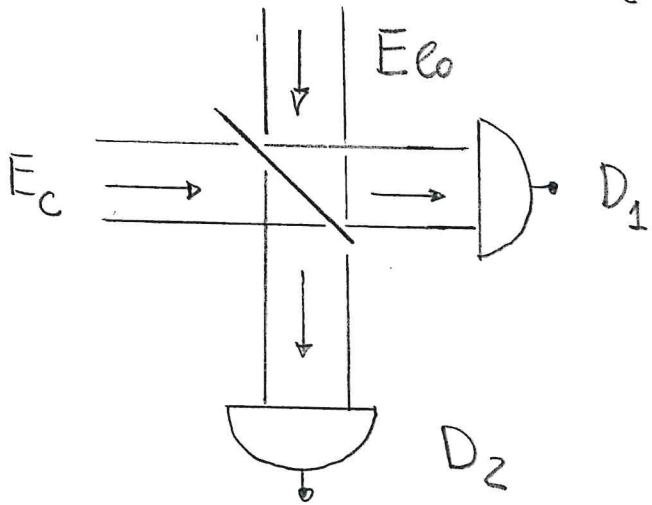
if $\varphi = \vartheta_0 + \pi/2 \rightarrow i(t) = R_\lambda \sqrt{P_c P_{lo}} \sin[\psi(t)]$

Heterodyne $\rightarrow \omega_{lo} \neq \omega_c \rightarrow i(t) = R_\lambda \sqrt{P_c P_{lo}} \cos[(\omega_c - \omega_{lo}) t + \vartheta(t) - \varphi]$

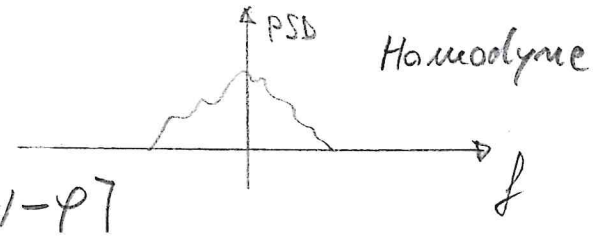
PCM ↓ Pulse code modulation

IM/DD ↑ intensity modul. Direct detection





optical mixer



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